

Power Flow Solver Using Numerical Methods

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Project Description

This project focuses on developing a computational tool to analyze power flow in an electrical power system. Power flow analysis is essential for determining the voltage magnitude, phase angles, and power distribution across a network of interconnected buses.

The motivation for this project comes from its importance in power system operation and planning, where engineers must ensure that electrical networks operate efficiently and within safe limits. This project aligns closely with real-world applications in power engineering and provides practical experience with numerical computation.

The program will use the Newton–Raphson method to solve the nonlinear algebraic equations that arise in power flow analysis. This method is widely used due to its fast convergence properties and accuracy when solving large systems of equations. The implementation will involve iterative computation of bus voltages until is satisfied with $\epsilon = 10^{-4}$ tolerance. [1]

Newton-Raphson Method Process

$$\mathbf{x} = \begin{bmatrix} \delta \\ \mathbf{V} \end{bmatrix} = \begin{bmatrix} \delta_2 \\ \vdots \\ \delta_N \\ V_2 \\ \vdots \\ V_N \end{bmatrix};$$

$$\mathbf{y} = \begin{bmatrix} \mathbf{P} \\ \mathbf{Q} \end{bmatrix} = \begin{bmatrix} P_2 \\ \vdots \\ P_N \\ Q_2 \\ \vdots \\ Q_N \end{bmatrix}$$

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} \mathbf{P}(\mathbf{x}) \\ \mathbf{Q}(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} P_2(\mathbf{x}) \\ \vdots \\ P_N(\mathbf{x}) \\ Q_2(\mathbf{x}) \\ \vdots \\ Q_N(\mathbf{x}) \end{bmatrix}$$

In each iteration i :

Step 1 Calculate the power mismatch

$$\Delta \mathbf{y}(i) = \begin{bmatrix} \Delta \mathbf{P}(i) \\ \Delta \mathbf{Q}(i) \end{bmatrix} = \begin{bmatrix} \mathbf{P} - \mathbf{P}[\mathbf{x}(i)] \\ \mathbf{Q} - \mathbf{Q}[\mathbf{x}(i)] \end{bmatrix}$$

Step 2 Calculate the Jacobian Matrix

Step 3 Use Gauss elimination and back substitution to solve

$$\begin{bmatrix} \mathbf{J1}(i) & \mathbf{J2}(i) \\ \mathbf{J3}(i) & \mathbf{J4}(i) \end{bmatrix} \begin{bmatrix} \Delta \boldsymbol{\delta}(i) \\ \Delta \mathbf{V}(i) \end{bmatrix} = \begin{bmatrix} \Delta \mathbf{P}(i) \\ \Delta \mathbf{Q}(i) \end{bmatrix}$$

Step 4 Update $\mathbf{x}$

$$\mathbf{x}(i+1) = \begin{bmatrix} \boldsymbol{\delta}(i+1) \\ \mathbf{V}(i+1) \end{bmatrix} = \begin{bmatrix} \boldsymbol{\delta}(i) \\ \mathbf{V}(i) \end{bmatrix} + \begin{bmatrix} \Delta \boldsymbol{\delta}(i) \\ \Delta \mathbf{V}(i) \end{bmatrix}$$

Figure 1: High-level overview of the numerical method [3]

Project Goals

The main objectives of this project are:

- (a) Develop a program capable of solving power flow problems for small to medium-sized power systems.
- (b) Implement the Newton–Raphson method to solve nonlinear equations efficiently.
- (c) Allow users to input system data such as bus types (PQ, PV, Slack), loads, generation, and line impedance.
- (d) Compute and display voltage magnitudes and phase angles at each bus.

Expected functionality

When the user enters the system information through a JSON file, the program calculates the voltage magnitude at each bus and the power flow in the network. If load values are changed, the program updates the results. If higher accuracy is requested, the program repeats its calculations more times until the results are precise enough. [2]

Requirements

The success of this project will be evaluated based on the following criteria:

- (a) The solver must compute bus voltage magnitudes and phase angles with a small error tolerance.
- (b) The program must accept number of buses, bus types, load and generation values, and line impedance parameters.
- (c) The program must output real and reactive power at each bus, and power flow in each transmission line.
- (d) The program must detect and handle invalid inputs (e.g., missing data, non-converging systems) and provide meaningful error messages.
- (e) The code must be modular, with separate functions for input processing and Jacobian matrix construction.

References

- [1] J. Duncan Glover, Thomas J. Overbye, and Mulukutla S. Sarma. *Power System Analysis and Design*. Cengage Learning, Boston, MA, 6th edition, 2017. Chapter 6, section 6.3.
- [2] OpenAI. Chatgpt (gpt-5.3 version). <https://chatgpt.com/share/69cd25c8-3b3c-83e8-a0cf-8bccd5963440>, 2026. Accessed: March 29, 2026.
- [3] Ying Zhang. Power flow 2 lecture notes. Course Handout, Oklahoma State University, January 2022. Course ECEN4293.